

WEEK 4 REPORT – POWER BI EDA

INTRODUCTION.

The Shopify stock dataset was used to study stock price and trading volume over time. The data was prepared and connected with a Date Table for time-based analysis. DAX measures were created to calculate moving averages and compare short-term and long-term trends.

DATA PREPARATION.

The Date field was used to organize the stock data according to different time periods. The data was arranged to analyze stock prices and trading volume using Power BI visuals and

DAX MEASURES.

DAX measures were created in Power BI to perform calculations on the stock data.

*20-Day Moving Average.

*50-Day Moving Average.

*Moving Average Comparison.

These measures were used to analyze moving-average trends in the Shopify stock data.

DAX Measure:

Volume 20 SMA =

VAR CurrentDate = MAX('Date Table'[Date])

VAR PeriodDays = 20

VAR DateInPeriod =

DATESINPERIOD(

'Date Table'[Date],

CurrentDate,

-PeriodDays,

DAY

)

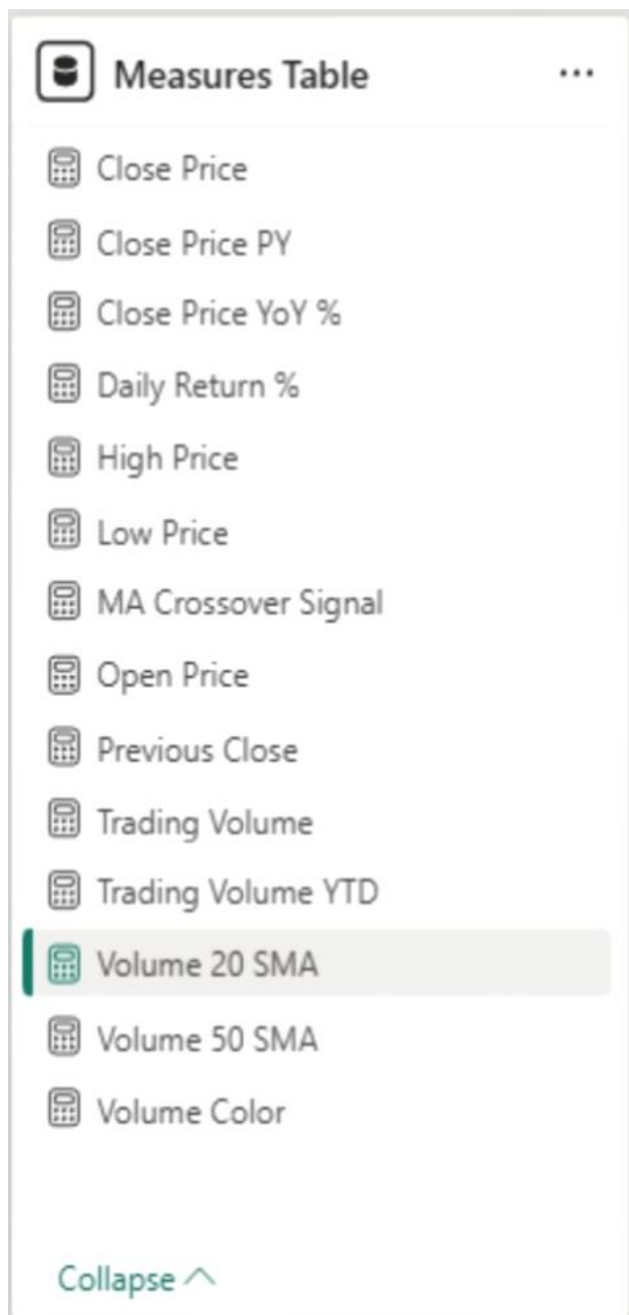
RETURN

CALCULATE(

AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume]),

DateInPeriod

)



50-Day Simple Moving Average:

A 50-Day Simple Moving Average (50 SMA) was created to provide a longer-term comparison with the 20-day average.

DAX Measure;

Volume 50 SMA =

VAR CurrentDate = MAX('Date Table'[Date])

VAR PeriodDays = 50

VAR DateInPeriod =

DATESINPERIOD(

'Date Table'[Date],

CurrentDate,

-PeriodDays,

DAY

)

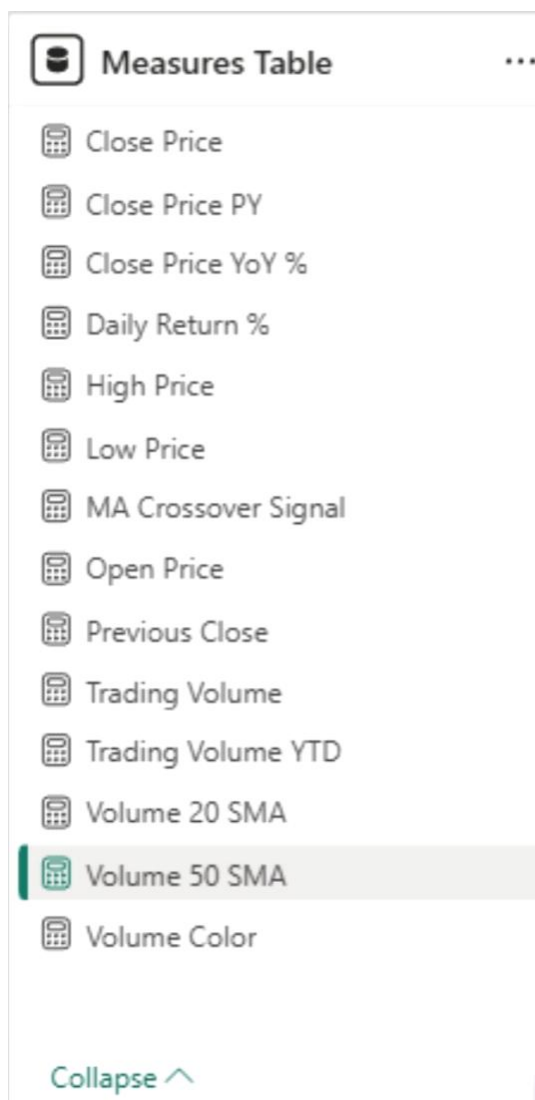
RETURN

CALCULATE(

AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume]),

DateInPeriod

)



Moving Average Crossover Signal.

A Moving Average Crossover Signal was created by comparing the 20 SMA and 50 SMA.

DAX Measure

MA Crossover Signal =

IF(

ISBLANK([Volume 20 SMA]) ||

ISBLANK([Volume 50 SMA]),

BLANK(),

IF(

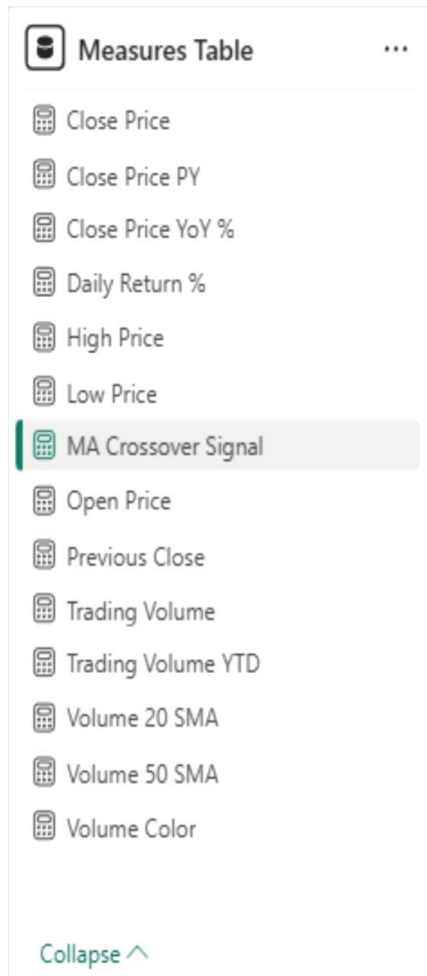
[Volume 20 SMA] > [Volume 50 SMA],

"Bullish (20 > 50)",

"Bearish (20 < 50)"

)

)



TRADING VOLUME ANALYSIS.

Trading volume was examined using Power BI line charts. The 20-Day SMA and 50-Day SMA were included to understand the movement of trading activity. The visualization helps identify periods where trading volume increased or decreased.

Visual Used:

*Chart: Line Chart

*X-Axis: Date

*Values: Volume, 20-Day SMA, 50-Day SMA

This chart provides a clear view of changes in trading activity over time.



STOCK PRICE ANALYSIS.

The Close Price was used to observe how the Shopify stock changed during the selected period. Daily Return % was also considered to understand the daily change in stock performance.

The analysis included:

- *High price.
- *Low price.
- *Open price.
- *Close price.

Metric Selector 

☐ Close Price

☐ High Price

☐ Low Price

☐ Open Price

HIGH AND LOW PRICE COMPARISON.

A line chart was created to compare the High Price and Low Price of the stock.

Visual Configuration;

- *Chart: Line Chart
- *X-Axis: Date
- *Values: High Price, Low Price

This visualization helps to observe the difference between the highest and lowest prices on different dates.



OPEN AND CLOSE PRICE COMPARISON.

Another line chart was created to compare the opening and closing prices.

Visual Configuration:

*Chart: Line Chart

*X-Axis: Date

*Values: Open Price, Close Price

This chart makes it easier to observe how the stock opened and closed during different trading periods.



SLICER AND FILTERING.

* A Month slicer was added to the Power BI dashboard. The slicer allows the user to select a particular month and view the related stock information.

* For example, when a month is selected, the charts automatically update according to the selected period. This makes the dashboard more interactive and helps users focus on a specific time period.

CUSTOM TOOLTIP.

A custom tooltip page named Stock_Tooltip was created. The tooltip provides additional information when the user moves the mouse pointer over a chart.

The tooltip contains:

*Open Price

*High Price

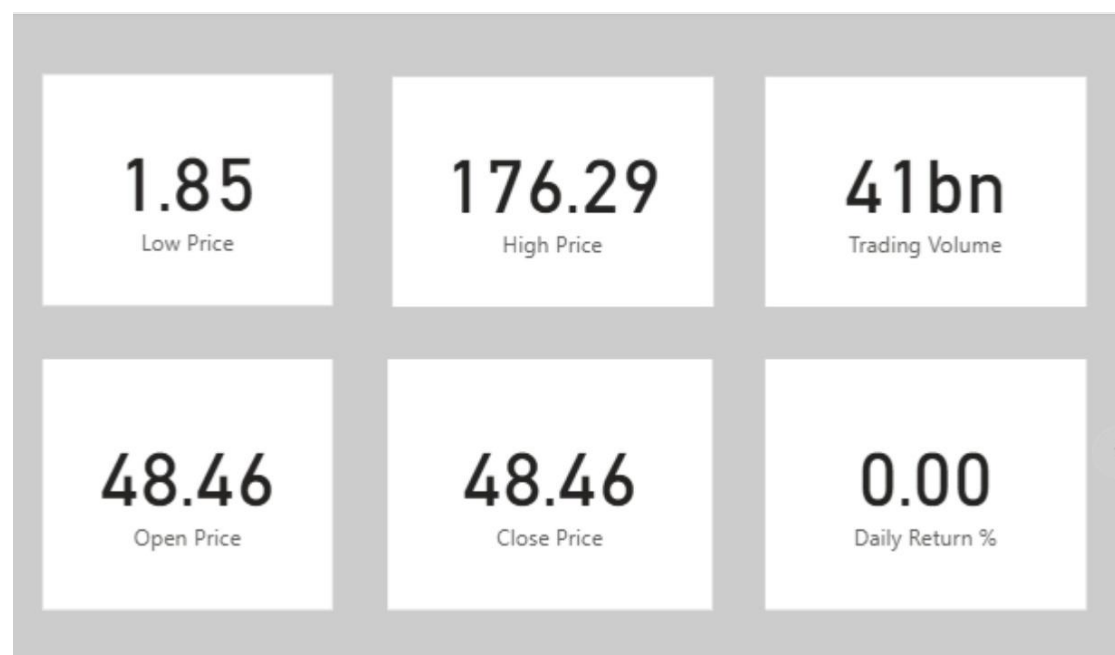
*Low Price

*Close Price

*Trading Volume

*Daily Change %

This feature provides more information without making the main dashboard crowded.



CONCLUSION.

The Week 4 Power BI EDA activity helped in understanding Shopify stock data through interactive visualizations and DAX calculations. Moving averages were used to study trading volume trends, while different line charts were used to examine stock prices.

The addition of a Month slicer and custom tooltip improved the interactivity of the dashboard. Overall, the analysis provided a simple and effective way to explore stock price and trading volume data using Power BI.

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